

S/N	Question	Options	ANS
1	Which of the following is NOT a primary reason for demanding high software quality?	A) Customer satisfaction B) Lower maintenance cost C) Faster hardware upgrades D) Compliance with standards	C
2	High-quality software directly improves _____ of a product in the market.	A) Price B) Reputation C) Weight D) Color	B
3	Poor software quality most often leads to an increase in _____.	A) Development speed B) Bug-fix effort C) User adoption D) Code readability	B
4	Which metric best reflects the importance of quality for safety-critical systems?	A) Lines of code B) Mean Time Between Failures (MTBF) C) Number of screens D) Story points	B
5	The “cost of quality” concept states that _____.	A) Investing in quality early reduces total cost B) Quality only adds cost C) Quality can be bought off-the-shelf D) Quality is unrelated to cost	A
6	Which attribute describes a system that can operate correctly under unexpected inputs?	A) Maintainability B) Robustness C) Portability D) Scalability	B
7	A software product that can be easily adapted to other platforms exhibits _____.	A) Reliability B) Usability C) Portability D) Testability	C
8	Which of these is not a quality attribute defined in ISO/IEC 25010?	A) Security B) Flexibility C) Efficiency D) Compatibility	B
9	“The ability of a component to be changed without affecting other components” is known as:	A) Modularity B) Reusability C) Maintainability D) Interoperability	C
10	The degree to which a system meets the needs of its users is called _____.	A) Functionality B) Suitability C) Usability D) Accuracy	C
11	Which phase follows Requirements Gathering in the classic Waterfall model?	A) Design B) Testing C) Deployment D) Maintenance	A
12	In Agile, the term “ Sprint Review ” belongs to which SDLC activity?	A) Requirements B) Implementation C) Testing D) Validation	D
13	Coding is primarily associated with which SDLC stage?	A) Planning B) Design C) Development D) Release	C
14	The “ Go-Live ” activity occurs during which phase?	A) Design B) Development C) Deployment D) Maintenance	C
15	Which SDLC phase produces the “ User Manual ”?	A) Requirements B) Design C) Testing D) Maintenance	D
16	Formal methods are primarily used to _____.	A) Speed up coding B) Provide mathematical specifications C) Reduce UI design effort D) Generate random test data	B
17	Which of the following is a formal specification language ?	A) UML B) HTML C) Z notation D) CSS	C

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18	Formal methods rely on which kind of reasoning?	A) Intuitive B) Empirical C) Deductive D) Inductive	C
19	A formal proof demonstrates that a system satisfies its specification _____.	A) Empirically B) By testing C) By logical deduction D) By user feedback	C
20	The term “formal” in formal methods refers to the use of _____.	A) Natural language B) Diagrams only C) Mathematical notation D) Color-coded comments	C
21	Which domain most often adopts formal methods?	A) E-commerce sites B) Safety-critical systems C) Social media apps D) Blogging platforms	B
22	Formal methods can be applied during _____ to detect design errors early.	A) Coding B) Deployment C) Requirements analysis D) Maintenance	C
23	Model checking is a technique used to _____.	A) Generate UI mock-ups B) Verify state-space properties C) Write user stories D) Perform load testing	B
24	Which of the following is a benefit of using formal methods?	A) Eliminates need for testing B) Guarantees zero defects C) Provides rigorous correctness evidence D) Shortens project schedule dramatically	C
25	Formal methods are least useful for which type of project?	A) Avionics control software B) Real-time embedded systems C) Simple command-line utilities D) Cryptographic protocol design	C
26	Formal specification can improve _____ among stakeholders.	A) Code size B) Communication clarity C) Database indexing D) Color scheme	B
27	Which activity is NOT a typical output of using formal methods?	A) Proof obligations B) Test scripts C) Formal model D) Invariant definitions	B
28	In software verification, a theorem prover is used to _____.	A) Compile code B) Prove logical assertions C) Generate UI screens D) Optimize memory usage	B
29	Formal methods can help certify compliance with standards such as _____.	A) ISO 9001 B) DO-178C C) GDPR D) PCI-DSS	B
30	When a system is modeled in CSP , what is primarily being described?	A) Data structures B) Communication processes C) Graphic layout D) Database schema	B